

ON THE MATRIX DIFFERENTIATION OF THE CHARACTERISTIC ROOTS OF MATRICES

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Summary

This paper describes properties of matrix derivatives (Bargmann (1964)) of characteristic roots of square non-singular real matrices $A(p \times p)$. Definitions and notations are introduced in Section 1. The proof of a Theorem for the derivative of a characteristic root is presented in section 2. Section 3 studies the relationship to derivatives of elementary symmetric functions (Nel (1972) and de Waal (1973)). Use is made of the theorems to shorten proofs of some well-known results in Section 4.

1. Introduction

The elements of square real-valued non-singular matrices are considered as real-valued numbers over which scalar functions, notably characteristic roots and elementary symmetric functions are defined. The matrix whose (i,j) -element is the partial derivative of the function with respect to the element a_{ij} ; formally

$$\frac{\partial f(A)}{\partial A} = B \Rightarrow b_{ij} = \frac{\partial f(A)}{\partial a_{ij}}$$

The rules governing such derivatives have considerably simplified proofs and exposition (Bargmann (1964), Neudecker (1967), Tracy and Dwyer (1969)).

The following notation will be used:

Capital letters denote matrices; transposition is indicated by prime (''); underlined lower-case letters denote column vectors if unprimed, and row vectors, if primed. A

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determinant is expressed as a matrix enclosed in vertical bars.

2. Matrix derivatives of characteristic roots

Neudecker (1967) derived an expression for the matrix derivative of the characteristic root λ_i of the matrix A , for the case of non-repeated roots e.g.

$$(2.1.1) \quad \frac{\partial \lambda_i}{\partial A} = \underline{v}_i \underline{u}_i'$$

where $\underline{v}_i$ and $\underline{u}_i$ are respectively the left and right characteristic vector of A with respect to λ_i and $\underline{v}_i$ and $\underline{u}_i$ such that

$$\underline{v}_i' \underline{u}_i = 1$$

Another formula for $\frac{\partial \lambda_i}{\partial A}$ can be derived by the following approach.

Theorem 2.1

$$(2.1.2) \quad \frac{\partial \lambda_i}{\partial A} = \prod_{j \neq i} (A' - \lambda_j I_p) / \prod_{j \neq i} (\lambda_i - \lambda_j)$$

Proof.

For the proof of (2.1.2) note that

$$\frac{\partial \operatorname{tr} A^r}{\partial a_{ij}} = \operatorname{tr} \left[\frac{\partial \operatorname{tr} A^r}{\partial A^r} \cdot \frac{\partial (A')^r}{\partial a_{ij}} \right] \quad (\text{Bargmann (1964); Chain Rule No. 2})$$

$$= \operatorname{tr} [I_p \cdot r(A')^{r-1} \frac{\partial A'}{\partial a_{ij}}]$$

$$= \operatorname{tr} [r(E\delta)_{ji} (A')^{r-1}] = [r(A')^{r-1}]_{ij}^*$$

$$\frac{\partial \operatorname{tr} A^r}{\partial A} = r(A')^{r-1}$$

$$\text{But, also } \frac{\partial}{\partial A} [\operatorname{tr} A^r] = \frac{\partial}{\partial A} \sum_{i=1}^p \lambda_i^r$$

* where $(A')_{ji}^{r-1}$ denotes the (i,j) element of $(A')^r$.

$$= r \sum_{i=1}^p \lambda_i^{r-1} \frac{\partial \lambda_i}{\partial A}$$

Thus

$$\sum_{i=1}^p \lambda_i^{r-1} \frac{\partial \lambda_i}{\partial A} = (A')^{r-1} \quad r=1, 2, \dots, p.$$

where A is of order and rank p by p.

In matrix form, this can be written as

$$(2.1.3) \quad \begin{bmatrix} I_p & I_p & \dots & I_p \\ \lambda_1 I_p & \lambda_2 I_p & \dots & \lambda_p I_p \\ \lambda_1^{p-1} I_p & \lambda_2^{p-1} I_p & \dots & \lambda_p^{p-1} I_p \end{bmatrix} \begin{bmatrix} \frac{\partial \lambda_1}{\partial A} \\ \frac{\partial \lambda_2}{\partial A} \\ \dots \\ \frac{\partial \lambda_p}{\partial A} \end{bmatrix} = \begin{bmatrix} I_p \\ A' \\ \dots \\ (A')^{p-1} \end{bmatrix}$$

The solution of this equation is similar to the solution of:

$$(2.1.4) \quad \begin{bmatrix} 1 & 1 & \dots & 1 \\ \lambda_1 & \lambda_2 & \dots & \lambda_p \\ \lambda_1^2 & \lambda_2^2 & \dots & \lambda_p^2 \\ \dots & \dots & \dots & \dots \\ \lambda_1^{p-1} & \lambda_2^{p-1} & \dots & \lambda_p^{p-1} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \dots \\ w_p \end{bmatrix} = \begin{bmatrix} 1 \\ x \\ x^2 \\ \dots \\ x^{p-1} \end{bmatrix}$$

which is the Vandermonde equation of the Lagrange interpolation at x through the points $\lambda_1, \lambda_2, \dots, \lambda_p$ and has the well-known solution:

$$(2.1.5) \quad w_i = \prod_{j \neq i} (x - \lambda_j) / \prod_{j \neq i} (\lambda_i - \lambda_j) \quad i=1, 2, \dots, p$$

Thus the solution of (2.1.3) is:

$$\frac{\partial \lambda_i}{\partial A} = \prod_{j \neq i} (A' - \lambda_j I_p) / \prod_{j \neq i} (\lambda_i - \lambda_j) \quad i=1, 2, \dots, p$$

Q.E.D.

Notice that the derivative of the characteristic root λ_i is a singular matrix and has the interesting property that:

$$(2.2) \quad (A' - \lambda_i I_p) \frac{\partial \lambda_i}{\partial A} = 0 \quad (p \times p)$$

which follows from the Cayley-Hamilton fundamental theorem (Finkbeiner (1960), p. 149).

If γ_i ($i=1, \dots, p$) denotes the characteristic roots of the matrix A^{-1} then by using Chain Rule No. 2 again:

$$(2.3) \quad \frac{\partial \lambda_i}{\partial A} = -A^{-T} \left\{ \prod_{j \neq i} (A^{-T} - \gamma_j I_p) / \prod_{j \neq i} (\gamma_i - \gamma_j) \right\} A^{-T} \quad \text{where } A^{-T} = (A')^{-1}$$

But $\gamma_i = 1/\lambda_i$ ($i=1, 2, \dots, p$)

so that the matrix derivatives of the reciprocals of the characteristic roots of A are

$$(2.4) \quad \frac{\partial \lambda_i^{-1}}{\partial A} = -A^{-T} \left\{ \prod_{j \neq i} (A^{-T} - \gamma_j I_p) / \prod_{j \neq i} (\lambda_j A^{-T} - I_p) / \prod_{j \neq i} (\lambda_j - \lambda_i) \right\} A^{-T}$$

Another interesting case is where the matrix A has rank one i.e. if $A = \underline{u} \underline{u}'$ where $\underline{u}$ ($p \times 1$). The only non-zero root of $|\underline{u} \underline{u}' - \lambda I| = 0$ is $\lambda_1 = \underline{u}' \underline{u}$. From (2.1.2) we have

$$(2.5) \quad \frac{\partial \lambda_1}{\partial (\underline{u} \underline{u}')} = \frac{(\underline{u} \underline{u}')^{p-1}}{(\underline{u}' \underline{u})^{p-1}} = \frac{\underline{u} \underline{u}'}{\underline{u}' \underline{u}} = \frac{1}{\lambda_1} \underline{u} \underline{u}'$$

3. Some results concerning B-polynomials

Let $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_p$ denote the characteristic roots of a matrix A ($p \times p$) then the j -th elementary symmetric function in of A can be written as:

$$\text{tr}_j A = \sum_{1 \leq i_1 < i_2 < \dots < i_j \leq p} \lambda_{i_1} \lambda_{i_2} \dots \lambda_{i_j}$$

which will also be denoted as a_j .

Definition

Let A be a $p \times p$ matrix, then the j -th B-polynomial of A , in argument x (which can be a scalar or a ($p \times p$) matrix variable) is defined as:

$$(3.1) \quad B_j(x) = \sum_{k=0}^j (-1)^k x^{j-k} \text{tr}_k A \text{ where } j=0,1,2,\dots,p$$

If x is a scalar variable then $x^0=1$ and if it is a $(p \times p)$ matrix variable then $x^0=I_p$. Further denote $\text{tr}_0 A=1$.

Lemma 3.1 (Nel (1972))

For any scalar λ and nonsingular matrix $A(p \times p)$:

$$(3.1.1) \quad \text{Adj}(I_p - \lambda A) = \sum_{j=0}^{p-1} \lambda^j B_j(A)$$

where $B_j(A)$ is the j -th B -polynomial with matrix argument A .

From this result the matrix derivatives of the elementary symmetric functions of A were derived as:

Lemma 3.2 (de Waal (1973), Nel (1972))

$$(3.2.1) \quad \frac{\partial \text{tr}_j A}{\partial A} = (-1)^{j-1} B_{j-1}(A'), \quad j=1,\dots,p$$

By using (3.2.1) another method to find the derivatives of the characteristic roots of A was obtained. This method yields some pleasant and interesting results regarding the B -polynomials.

From (3.2.1) it follows:

$$\frac{\partial a_1}{\partial A} = \frac{\partial \lambda_1}{\partial A} + \frac{\partial \lambda_2}{\partial A} \dots + \frac{\partial \lambda_p}{\partial A} = B_0(A')$$

$$\frac{\partial a_2}{\partial A} = \sum_{i \neq 1} \lambda_i \frac{\partial \lambda_1}{\partial A} + \sum_{i \neq 2} \lambda_i \frac{\partial \lambda_2}{\partial A} + \dots + \sum_{i \neq p} \lambda_i \frac{\partial \lambda_p}{\partial A} = -B_1(A')$$

$$\frac{\partial a_p}{\partial A} = \prod_{i \neq 1} \lambda_i \frac{\partial \lambda_i}{\partial A} + \prod_{i \neq 2} \lambda_i \frac{\partial \lambda_2}{\partial A} + \dots + \prod_{i \neq p} \lambda_i \frac{\partial \lambda_p}{\partial A} = (-1)^{p-1} B_{p-1}(A')$$

Keeping in mind that: $\sum_{i \neq j} \lambda_i = B_1(\lambda_j)$, $\sum_{i \neq j \neq k} \lambda_i \lambda_k = B_2(\lambda_j)$, etc. these equations can be

written in matrix form as.

$$\begin{bmatrix} B_0(\lambda_1)I_p & B_0(\lambda_2)I_p & \dots & B_0(\lambda_p)I_p \\ B_1(\lambda_1)I_p & B_1(\lambda_2)I_p & \dots & B_1(\lambda_p)I_p \\ \vdots & \vdots & \ddots & \vdots \\ B_{p-1}(\lambda_1)I_p & B_{p-1}(\lambda_2)I_p & \dots & B_{p-1}(\lambda_p)I_p \end{bmatrix} \begin{bmatrix} \frac{\partial \lambda_1}{\partial A} \\ \frac{\partial \lambda_2}{\partial A} \\ \vdots \\ \frac{\partial \lambda_p}{\partial A} \end{bmatrix} = \begin{bmatrix} B_0(A') \\ B_1(A') \\ \vdots \\ B_{p-1}(A') \end{bmatrix}$$

which in turn can be written as a matrix product:

(3.3)

$$\begin{bmatrix} I_p & 0 & \dots & 0 & 0 \\ -a_1 I_p & I_p & \dots & 0 & 0 \\ a_2 I_p & -a_1 I_p & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ (-1)^{p-1} a_{p-1} I_p & (-1)^{p-2} a_{p-2} I_p & \dots & -a_1 I_p & I_p \end{bmatrix} \begin{bmatrix} I_p & \dots & I_p \\ \lambda_1 I_p & \dots & \lambda_p I_p \\ \lambda_1^2 I_p & \dots & \lambda_p^2 I_p \\ \vdots & \ddots & \vdots \\ \lambda_1^{p-1} I_p & \dots & \lambda_p^{p-1} I_p \end{bmatrix} \begin{bmatrix} \frac{\partial \lambda_1}{\partial A} \\ \frac{\partial \lambda_2}{\partial A} \\ \frac{\partial \lambda_3}{\partial A} \\ \vdots \\ \frac{\partial \lambda_p}{\partial A} \end{bmatrix} = \begin{bmatrix} B_0(A') \\ B_1(A') \\ B_2(A') \\ \vdots \\ B_{p-1}(A') \end{bmatrix}$$

This equation is of the form:

$$(3.4) \quad (W \otimes I_p)(V \otimes I_p) \begin{bmatrix} \frac{\partial \lambda_i}{\partial A} \end{bmatrix}_{p^2 \times p} = [B_{i-1}(A')]_{p^2 \times p}$$

where $A \otimes B$ denotes the Kronecker product of A and B and where V is a Vandermonde matrix in the characteristic roots of A .

By inverting the lower-triangular matrix $W(p \times p)$ a similar matrix W^{-1} with elements the homogeneous product sums of the latent roots of A is obtained. These homogeneous functions are also called h -functions (David and Kendall (1966)) and will be denoted by $h_i(A)$, $i=1,2,\dots,p$.

$$\text{Since } W^{-1} = \begin{bmatrix} 1 & 0 & \dots & 0 & 0 \\ h_1(A) & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ h_{p-1}(A) & h_{p-2}(A) & \dots & h_1(A) & 1 \end{bmatrix}$$

we obtain from 3.4

$$\begin{bmatrix} I_p & \dots & I_p \\ \lambda_1 I_p & \dots & \lambda_p I_p \\ \vdots & \ddots & \vdots \\ \lambda_1^{p-1} I_p & \dots & \lambda_p^{p-1} I_p \end{bmatrix} \begin{bmatrix} \frac{\partial \lambda_1}{\partial A} \\ \frac{\partial \lambda_2}{\partial A} \\ \vdots \\ \frac{\partial \lambda_p}{\partial A} \end{bmatrix} = \begin{bmatrix} I_p & 0 & \dots & 0 & B_0(A') \\ I_p h_1(A) & I_p & \dots & 0 & B_1(A') \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ I_p h_{p-1}(A) & I_p h_{p-2}(A) & \dots & I_p & B_{p-1}(A') \end{bmatrix}$$

which is essentially the same as (2.1.3). Thus we have proved that:

$$(3.6) \quad \sum_{k=0}^j h_{j-k}(A) B_k(x) = x^j$$

where $B_k(x)$ is defined on the same matrix A and $h_0(A) = 1$.

Some other interesting results concerning the B -polynomials are:

$$(3.7) \quad \begin{aligned} \text{Adj}(A) &= (-1)^{p-1} B_{p-1}(A) \\ &= (-1)^{p-1} (A^{p-1} - A^{p-2} \text{tr}_1 A + \dots + (-1)^{p-1} I_p \text{tr}_{p-1} A) \end{aligned}$$

The latter being well known.

By writing the product $\prod_{j \neq i} (A - \lambda_j I_p)$ first as a series in λ_i and then as a series in A we conclude:

$$(3.8) \quad \sum_{k=0}^{p-1} B_k(A) \lambda_i^{p-1-k} = \sum_{k=0}^{p-1} B_k(\lambda_i) A^{p-1-k}$$

If we use (3.8) together with the easily proven result

$$(3.9) \quad \sum_{k=0}^{p-1} B_k(\lambda_i) \lambda_i^{p-1-k} = \sum_{k=0}^{p-1} (-1)^k (p-k) \lambda_i^{p-1-k} \text{tr}_k A$$

the derivative of the i -th characteristic roots can also be written as:

$$(3.10) \quad \frac{\partial \lambda_i}{\partial A} = \frac{\sum_{k=0}^{p-1} B_k(A') \lambda_i^{p-1-k}}{\sum_{k=0}^{p-1} (-1)^k (p-k) \lambda_i^{p-1-k} \text{tr}_k A}$$

4. Use of formulae in derivations

Although the results (2.1.2) and (3.10) are useful for recognition purposes, the result (2.1.1) of Neudecker (1967) is easier to apply, and can simplify many of the existing proofs.

The best-known results of matrix differentiation are (e.g. Bargmann (1964))

$$(4.1) \quad \frac{\partial \text{tr} A}{\partial A} = I_p$$

and

$$(4.2) \quad \frac{\partial \log |A|}{\partial A} = A^{-T}$$

The usual proof of the latter is by Laplace development

$$\text{By (2.1.1): } \frac{\partial \text{tr} A}{\partial A} = \sum_{i=1}^p \frac{\partial \lambda_i}{\partial A} = \sum_{i=1}^p v_i u_i' = I_p$$

For other proofs note that if the characteristic vectors are all linearly independent, then:

$$V A U = U^{-1} A U = D = \text{diag}(\lambda_1, \dots, \lambda_p)$$

where the right eigenvectors (columns of U) are ordered in the same way as the associated roots.

Thus:

$$A = U D_\lambda U^{-1} = U D_\lambda V = \sum_{i=1}^p \lambda_i u_i v_i'$$

Then, for any polynomial in A:

$$P(A) = U D_{p(\lambda)} V = \sum_{i=1}^p p(\lambda_i) u_i v_i'$$

the usual result.

$$\begin{aligned} \text{Then } \frac{\partial \log |A|}{\partial A} &= \sum_{i=1}^p \frac{\partial \log \lambda_i}{\partial A} = \sum_{i=1}^p \frac{1}{\lambda_i} \frac{\partial \lambda_i}{\partial A} \\ &= \sum_{i=1}^p \frac{1}{\lambda_i} v_i u_i' = V' D_{1/\lambda} U' \end{aligned}$$

$$= (U D_{\lambda} V)^{-T} = A^{-T}$$

$$\begin{aligned} \text{and } \frac{\partial \operatorname{tr} A^r}{\partial A} &= r \sum_{i=1}^p \lambda_i^{r-1} \frac{\partial \lambda_i}{\partial A} \\ &= r \sum_{i=1}^p \lambda_i^{r-1} v_i u_i' = r(A')^{r-1} \end{aligned}$$

Compare this with the well-known result in Section 2.

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